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FINAL PROJECT REPORT

Project Title:

PREDICTING VEHICLE FUEL EFFICIENCY USING MACHINE LEARNING: A REGRESSION ANALYSIS OF THE AUTO MPG DATASET

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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Abstract

In this project I investigate how accurately automobile fuel efficiency, measured in miles per gallon (mpg), can be predicted from a small set of technical vehicle attributes using supervised machine learning. I work with the original Auto MPG dataset from the UCI Machine Learning Repository, which contains fuel-consumption and specification data for cars produced between 1970 and 1982. After loading the raw data, I correct the horsepower data type, remove six rows with missing values and confirm that 392 complete cases remain. I then carry out exploratory data analysis to examine the distributions of cylinders, displacement, horsepower, weight, acceleration, model year and origin, and to quantify their relationships with mpg. Guided by previous work on Auto MPG and fuel-efficiency prediction, I implement and compare Linear Regression, Ridge, Lasso, Decision Tree and Random Forest models using an 80/20 train–test split and 5-fold cross-validation. Model performance is evaluated using root mean squared error (RMSE), mean squared error (MSE) and coefficient of determination (R^2), supported by residual and actual-versus-predicted plots. The Random Forest Regressor emerges as the best-performing model, achieving a cross-validated RMSE of about 3.0 mpg and a test RMSE of approximately 2.3 mpg, with an R^2 close to 0.91, which is consistent with the performance reported in similar studies. Feature-importance analysis shows that vehicle weight and engine displacement are the dominant predictors of fuel efficiency, followed by horsepower and model year. I conclude that tree-based ensemble methods are well suited to this type of tabular automotive data and that they can provide practically useful fuel-efficiency estimates for new vehicle designs.

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Chapter 1

1. Introduction

Reducing fuel consumption and greenhouse-gas emissions from road transport remains a central goal for governments and manufacturers, driven by climate targets, fuel costs and increasingly strict emissions regulations (e.g. Chen et al., 2025;). At the same time, historical vehicle data linking technical specifications to observed fuel consumption is widely available. This creates an opportunity to use data-driven models to estimate how design choices such as engine displacement, power output and vehicle weight affect fuel efficiency, particularly at early stages of the design process when prototyping is expensive.

The Auto MPG dataset from the UCI Machine Learning Repository is a widely used benchmark for this problem (UCI, 2023). It contains fuel-consumption data and technical attributes for cars produced between 1970 and 1982, originally compiled for the StatLib library at Carnegie Mellon University and the 1983 American Statistical Association Exposition. Because it is relatively small yet structurally rich, the dataset has been used in classical regression analyses and modern machine-learning studies. Linear-regression-based work has shown that mpg is strongly related to weight, displacement and horsepower, but also highlighted multicollinearity and non-linear effects (Regressit, 2024). More recent studies report that tree-based ensemble methods such as Random Forest often outperform linear models on fuel-efficiency prediction tasks, including on Auto MPG itself (Rachida Sghir et al., 2019; Chen et al., 2025; El-Shoraa, 2023).

My project builds on this context and focuses on the following research question:

How accurately can fuel efficiency (mpg) be predicted from vehicle attributes such as cylinders, displacement, horsepower, weight, acceleration, model year and origin, and which regression model provides the best predictive performance on the Auto MPG dataset?

To address this question, I work directly with the original UCI data to maintain full control over data cleaning and pre-processing. I perform exploratory data analysis to understand distributions and correlations; for example, a Pearson correlation heatmap (Figure 3.1) shows that mpg is strongly negatively correlated with weight, displacement and horsepower, and positively correlated with model year, replicating patterns identified in earlier work (Regressit, 2024). On the modelling side, I follow a staged approach: I begin with ordinary Least Squares Linear Regression as a baseline, add Ridge and Lasso to address multicollinearity, then introduce a Decision Tree Regressor and finally a Random Forest Regressor as the main non-linear ensemble model (Rachida Sghir et al., 2019; Chen et al., 2025; El-Shoraa, 2023). All models are evaluated using an 80/20 train–test split, 5-fold cross-validation and multiple regression metrics, in line with the MSc Data Science Project Handbook and established methodological guidance (Irsic, 2025; TensorFlow, 2023).

Ethically, this project is straightforward. The Auto MPG dataset contains only technical information about vehicles and no personal identifiers (UCI, 2023). The data were collected and published for research and education, and I use them solely in that context. I do not collect any new data from human participants, and I store both the dataset and my code

securely in a private GitHub repository. Because I work exclusively with secondary, anonymised data, formal UH Ethics Committee approval is not required, although I follow the general ethical principles set out in the project handbook .

Chapter 2

2. Background and Literature Review

In this chapter I position my work within existing research on vehicle fuel-efficiency prediction, with a focus on studies that use the Auto MPG dataset or closely related data. My aim is to explain which models other authors have used for this problem, what results they reported, and how that influenced my own choices.

2.1 Linear regression on Auto MPG

The Auto MPG dataset has been analysed in depth using classical linear regression. A detailed treatment is the regression analysis and data-preparation note by Regressit.com, which examines how mpg depends on weight, displacement and horsepower using stepwise model selection and residual diagnostics (Regressit, 2024). They show that vehicle weight alone explains a large proportion of the variance in mpg and that adding displacement and horsepower improves fit but introduces strong multicollinearity among predictors. Their residual plots also highlight that the relationship between weight and mpg is not perfectly linear, with very heavy cars achieving particularly low mpg.

I used this work as a baseline reference for what a well-implemented linear model can achieve on this dataset. It justified my decision to start with an ordinary Least Squares Linear Regression model and treat it as a serious benchmark. It also motivated two aspects of my exploratory analysis: reproducing their focus on weight, displacement and horsepower via a correlation heatmap (Figure 3.1), and planning to inspect residual plots for my linear model. Because they emphasise multicollinearity, their paper also influenced my decision to include Ridge and Lasso regression, which are designed to stabilise coefficients when predictors are highly correlated.

2.2 Tree-based ensembles for fuel-efficiency prediction

Several authors have moved beyond pure linear models and explored tree-based ensembles. Rachida Sghir and colleagues maintain a GitHub repository applying a full machine-learning pipeline to the Auto MPG dataset, comparing Linear Regression with Random Forest and other algorithms (Rachida Sghir et al., 2019). Their results show that Random Forest achieves lower mean squared error than Linear Regression and that it identifies weight and displacement as the most important features, with horsepower and model year also contributing strongly. They also note that a single decision tree tends to overfit, while the Random Forest generalises better by averaging many trees.

I treated this repository as a practical demonstration of what can be achieved with ensemble methods on exactly my dataset. It encouraged me to include a Decision Tree Regressor as a non-linear baseline and to select Random Forest as my main candidate for the final model. When I later computed the feature-importance scores for my best Random Forest model, I observed the same pattern—weight and displacement dominating—echoing their findings.

Beyond Auto MPG, Chen et al. (2025) developed machine-learning models for vehicle fuel-efficiency prediction using a larger dataset with driving conditions as well as vehicle specifications (Chen et al., 2025). They compared several algorithms, including Random

Forest and gradient-boosted trees, and concluded that ensemble tree methods provided the best predictive performance and robustness. El-Shoraa (2023) applied Random Forest to an extended vehicle-efficiency dataset and again found that it outperformed both single decision trees and linear models, particularly when modelling non-linear interactions (El-Shoraa, 2023). I interpreted these results as a strong signal that tree-based ensembles are a natural choice for my project.

2.3 Educational use of Auto MPG and evaluation practices

The Auto MPG dataset is also widely used in teaching materials. The official TensorFlow tutorial on “Basic regression: Predict fuel efficiency” uses this dataset to demonstrate how to build and evaluate a regression model (TensorFlow, 2023). Although my project does not use neural networks, I adopted their evaluation practices: splitting the data into training and test sets, using RMSE and R^2 as metrics, and plotting residuals to understand model behaviour.

I also reviewed several Auto MPG analyses on Kaggle. These notebooks typically implement Linear Regression and sometimes Decision Trees or Random Forest, and they report that Random Forest tends to achieve slightly lower error than the linear models. Their reported RMSE values provided an informal sanity check: my Random Forest test RMSE of about 2.3 mpg lies in the same range, suggesting that my implementation behaves sensibly.

The MSc Data Science Project Handbook and project lectures stress the importance of rigorous evaluation and optimisation (Irsic, 2025). They recommend using more than one metric, performing hyperparameter tuning and demonstrating improvements over baseline models. This guidance influenced my choice to use GridSearchCV for Ridge, Lasso, Decision Tree and Random Forest, and to use 5-fold cross-validation plus a hold-out test set.

2.4 Summary and implications

The literature leads to three main conclusions that directly shape my methodology. First, classical linear regression studies show that mpg can be modelled reasonably well from a small set of technical variables, but they also reveal multicollinearity and non-linear relationships. This is why I start with Linear Regression, add Ridge and Lasso, and pay attention to correlation and residual plots. Second, multiple studies indicate that Random Forest and other tree-based ensembles usually outperform linear models and single decision trees on fuel-efficiency prediction tasks, including Auto MPG. This justifies my choice of Random Forest as the main candidate model and my decision to treat the single Decision Tree as a baseline rather than the final solution. Third, best-practice guidelines from tutorials and the project handbook highlight the need for careful evaluation using multiple metrics, cross-validation and hyperparameter tuning. In the next chapter I describe how I applied these insights in my own pipeline.

Chapter 3

3. Data and Methodology

In this chapter I describe the dataset I used, how I prepared it for analysis and the modelling pipeline I implemented. I focus on what I did, why I did it that way, and how each step connects to the research question.

3.1 Dataset description and origin

The dataset for this project is the Auto MPG dataset from the UCI Machine Learning Repository (UCI, 2023). It contains city-cycle fuel-consumption measurements for 398 passenger cars produced between model years 1970 and 1982, along with a set of technical attributes describing each vehicle. The data were originally compiled for the StatLib library at Carnegie Mellon University and used in the 1983 American Statistical Association Exposition before being hosted at UCI as a benchmark for regression and exploratory data analysis (UCI, 2023).

Each record corresponds to one car model and includes nine variables: mpg (miles per gallon), cylinders, displacement (cubic inches), horsepower, weight (pounds), acceleration (0–60 mph time in seconds), model_year (encoded as 70–82), origin (region code: 1, 2 or 3), and car_name (a text label). For my modelling, I treat mpg as the continuous target variable and use all numeric variables apart from car_name as candidate predictors. The car_name field is retained only as an identifier during exploration.

From a machine-learning perspective, the task is to learn a mapping from the predictor vector

x = (cylinders, displacement, horsepower, weight, acceleration, model_year, origin)

to the target y = mpg.

Once trained, such a model can be used to estimate mpg for new vehicle designs specified by the same attributes, which directly addresses my research question about predictive accuracy and model choice.

3.2 Data loading and cleaning

I started by loading the raw data file auto-mpg.data from the UCI site using pandas.read_csv, manually specifying the column names recommended in the UCI documentation (UCI, 2023). An initial call to df.head() and df.shape confirmed that I had 398 rows and 9 columns, matching the repository description. I then used df.info() to inspect data types and completeness. Most columns were already numeric, but horsepower and car_name appeared as object type. While car_name is textual, horsepower should be numeric, so I treated this as a data-quality issue.

The UCI documentation indicates that some horsepower values are recorded as “?” in the original file (UCI, 2023). To convert horsepower to a numeric type while dealing with these entries explicitly, I applied pd.to_numeric(df["horsepower"], errors="coerce"). This converts valid numeric strings to floats and set any “?” values to NaN. I then dropped rows with missing horsepower using df.dropna(subset=["horsepower"]). This reduced the dataset from 398 to 392 rows, meaning that six cars had incomplete horsepower information. Given this small number and the importance of horsepower as a predictor, I decided that dropping these rows was preferable to imputing potentially misleading values. After dropping, I recast horsepower to float64 and weight to int64 to ensure consistent numeric types.

To verify that the cleaning had worked, I reran df.info() and checked df.isna().sum(). All predictor and target columns had zero missing values and appropriate numeric types. At this point I had a clean dataset of 392 complete cases ready for exploratory analysis and modelling. Keeping the cleaning process simple and transparent was deliberate: I wanted to reproduce the behaviour reported in previous studies that use the original UCI data and to be able to explain every transformation I applied.

3.3 Exploratory data analysis

With a clean dataset, I conducted an exploratory data analysis (EDA) to understand the distributions of individual variables and the relationships between predictors and mpg. My first step was to generate boxplots for all numeric variables. These showed that mpg values are concentrated between roughly 15 and 30 mpg, with a handful of high-efficiency cars above 35 mpg, while displacement and weight have long upper tails due to large, heavy cars. Horsepower and acceleration also showed several extreme values. These patterns suggested that my models would need to cope with skewed distributions and a few outliers rather than perfectly symmetric data.

I then examined the discrete variables origin, cylinders and model_year using histograms and, for origin, a pie chart. I observed that most cars come from origin 1, with smaller but substantial groups from origins 2 and 3. Four-cylinder cars dominate the cylinders distribution, followed by eight-cylinder and six-cylinder models. The model_year histogram shows that vehicles are fairly evenly spread across 1970–1982, with some years more heavily represented. These plots reassured me that the dataset covers a mix of economy and performance vehicles across a meaningful time span, but they also highlighted that origin is imbalanced.

To study pairwise relationships, I created scatter plots of weight versus mpg, weight versus horsepower, and model_year versus mpg. The weight–mpg plot shows a clear downward trend: heavier cars tend to have much lower mpg. The weight–horsepower plot confirms that heavier cars usually have more powerful engines. The model_year–mpg plot shows that average mpg increases over time, consistent with technological improvements and regulatory changes during the 1970s and early 1980s (Chen et al., 2025).

To quantify these patterns, I computed the Pearson correlation matrix for mpg and the numeric predictors and visualised it as a heatmap (Figure 3.1). This confirmed that mpg is strongly negatively correlated with cylinders, displacement, horsepower and weight, and positively correlated with model_year and origin. It also revealed strong positive correlations

among the engine and weight variables themselves, especially between displacement and weight and between displacement and horsepower. This multicollinearity matches what earlier linear-regression studies report and reinforced my decision to include Ridge and Lasso regression in addition to ordinary Linear Regression (Regressit, 2024). Kernel-density plots for mpg, horsepower, displacement and acceleration showed that horsepower and displacement are right-skewed, acceleration is roughly bell-shaped, and mpg is slightly right-skewed, confirming that the dataset is realistic and not perfectly Gaussian.

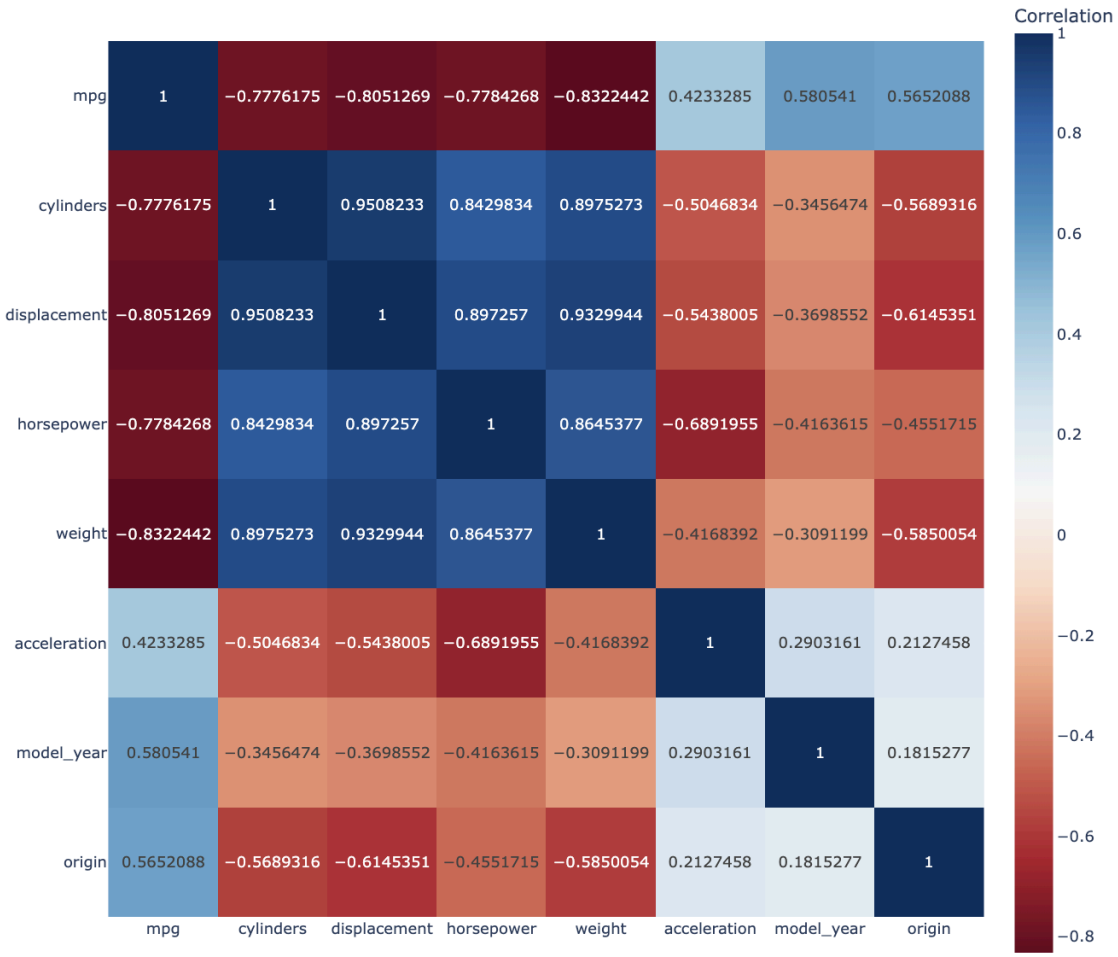

Figure 3.1: Pearson correlation heatmap for mpg and key predictors (cylinders, displacement, horsepower, weight, acceleration, model_year and origin). The strong negative correlations between mpg and the engine/weight variables, and the positive correlation with model_year, match patterns reported in earlier Auto MPG regression studies (Regressit, 2024).

3.4 Feature selection and train–test split

Based on the EDA and the literature, I used mpg as the target variable and treated cylinders, displacement, horsepower, weight, acceleration, model_year and origin as the main predictors. I excluded car_name from the modelling because it is a high-cardinality text label that would require complex encoding and would not generalise well beyond the specific models in the dataset.

For most experiments I used a “full” feature set containing all seven numeric predictors. To explore the contribution of temporal and regional context, I also defined a reduced feature set that excluded model_year and origin, keeping only cylinders, displacement, horsepower, weight and acceleration. This allowed me to test how much predictive power is gained by including information about when and where the car was built.

To evaluate the models, I split the data into training and test sets using train_test_split from scikit-learn, with 80% of the data used for training and 20% for testing and a fixed random seed for reproducibility. This resulted in 313 training instances and 79 test instances. The training set was used for cross-validation and hyperparameter tuning, while the test set was held back until the final evaluation of each model to obtain an unbiased estimate of performance on unseen vehicles.

3.5 Modelling strategy and hyperparameter tuning

My modelling strategy was shaped directly by the literature review. I wanted to see how far simple linear models could go and then to test whether tree-based ensembles could improve on them.

I began with an ordinary Least Squares Linear Regression model using the full feature set. I evaluated this baseline using 5-fold cross-validation on the training data, with RMSE and R^2 as the main metrics. RMSE, measured in mpg, gives a direct sense of the average prediction error, while R^2 indicates the proportion of variance in mpg explained by the model. I chose RMSE rather than mean absolute error (MAE) because RMSE penalises larger errors more strongly, which is desirable in this context: substantially mis-predicting mpg for some cars would be more problematic than many small deviations, and RMSE is still expressed in the original mpg units.

Because of the multicollinearity observed in the correlation heatmap, I next implemented Ridge and Lasso regression and tuned the regularisation strength using GridSearchCV with 5-fold cross-validation, using negative RMSE as the scoring metric (Regressit, 2024). This allowed me to test whether regularisation produced a measurable improvement in generalisation compared with ordinary Linear Regression.

To explore non-linear relationships and interactions, I then trained a Decision Tree Regressor. I chose this model as a simple non-linear baseline and as a stepping stone towards Random Forest. I tuned the maximum depth and minimum samples per leaf using GridSearchCV, again with 5-fold cross-validation and RMSE as the primary metric.

Finally, guided by studies reporting strong performance of tree-based ensembles on fuel-efficiency tasks, I implemented a Random Forest Regressor (Rachida Sghir et al., 2019; Chen et al., 2025; El-Shoraa, 2023). I first ran a coarse grid over the number of trees, maximum depth and minimum samples per leaf, and then refined the search to include different values of the max_features parameter. All Random Forest models were evaluated using 5-fold cross-validation on the training set. Once I identified the best configuration, I retrained it on the full training data and evaluated it on the held-out test set, computing MSE, RMSE and R^2 and producing residual and actual-versus-predicted plots.

Throughout the modelling process I also ran a sensitivity check by fitting a linear model on the reduced feature set without model_year and origin. Comparing its performance with the full-feature linear model allowed me to quantify the value of including temporal and regional information. For the final Random Forest model I computed feature-importance scores, which I later used to interpret the relative influence of weight, displacement, horsepower, model_year and the other predictors.

Chapter 4

4. Results

In this chapter I present the empirical results of my analysis and modelling and relate them back to the research question. I begin with the main findings from the exploratory analysis, then report the performance of the linear and regularised models, followed by the tree-based methods and the final tuned Random Forest model.

4.1 Exploratory analysis findings

The exploratory data analysis confirmed several intuitive relationships and quantified their strength. The scatter plot of mpg versus weight (Figure 4.1) shows a clear downward trend: heavier cars consistently achieve lower fuel efficiency. The points are not perfectly aligned on a straight line, but the negative pattern is unmistakable and becomes more pronounced for very heavy vehicles. This visual impression matches the strong negative correlation between weight and mpg in the correlation heatmap.

The plot of average mpg versus model_year (Figure 4.2) reveals a steady increase in fuel efficiency over time. Cars from the early 1970s cluster at lower mpg values, while models from the late 1970s and early 1980s achieve much higher mpg. This pattern is consistent with external knowledge about tightening emissions regulations and the oil crises of the 1970s, which pushed manufacturers towards more economical designs (Chen et al., 2025). It also justifies the inclusion of model_year as a predictor: when a car was built clearly influences its likely fuel efficiency.

The correlation heatmap (Figure 3.1) provides a compact summary of these relationships. Mpg is strongly negatively correlated with cylinders, displacement, horsepower and weight, and positively correlated with model_year and origin. At the same time, the engine-size variables themselves are highly positively correlated with each other. This confirmed that I was dealing with a multicollinear set of predictors and suggested that regularised linear models and tree-based models might behave differently from ordinary least squares (Regressit, 2024).

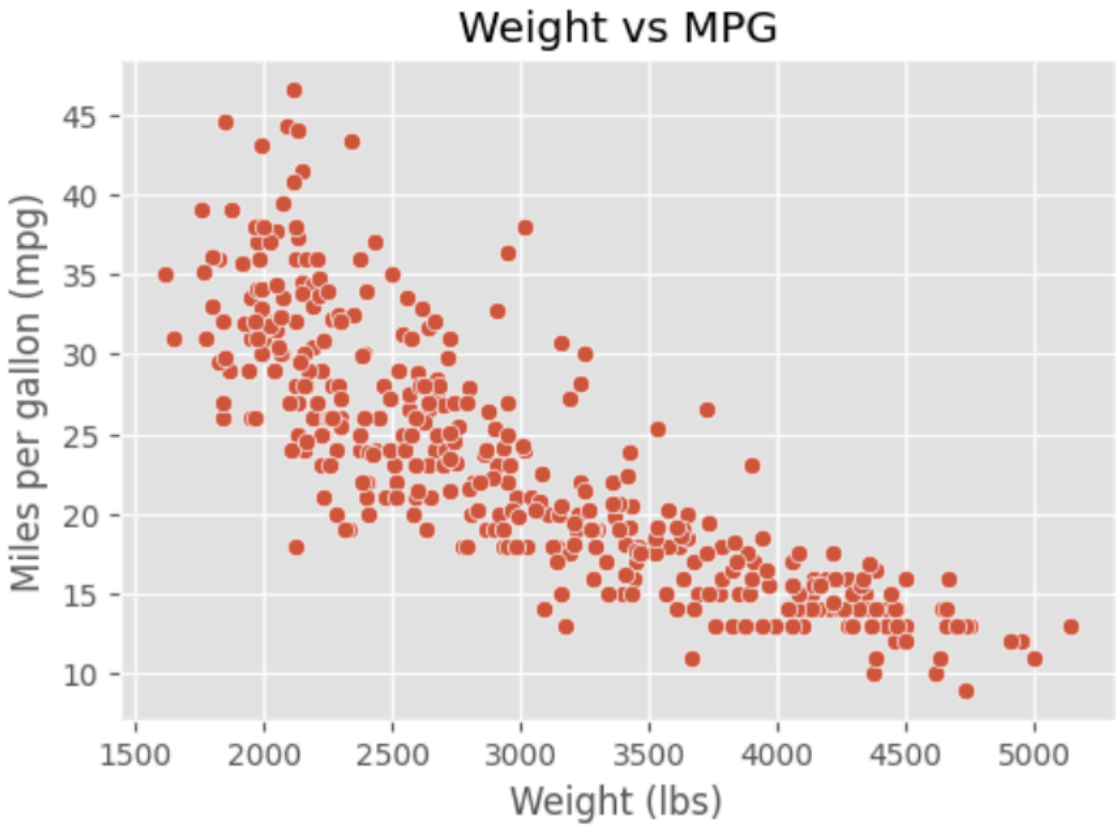

Figure 4.1: Scatter plot of mpg versus weight. Heavier vehicles consistently achieve lower mpg, with a strong negative trend that becomes more pronounced for very heavy cars.

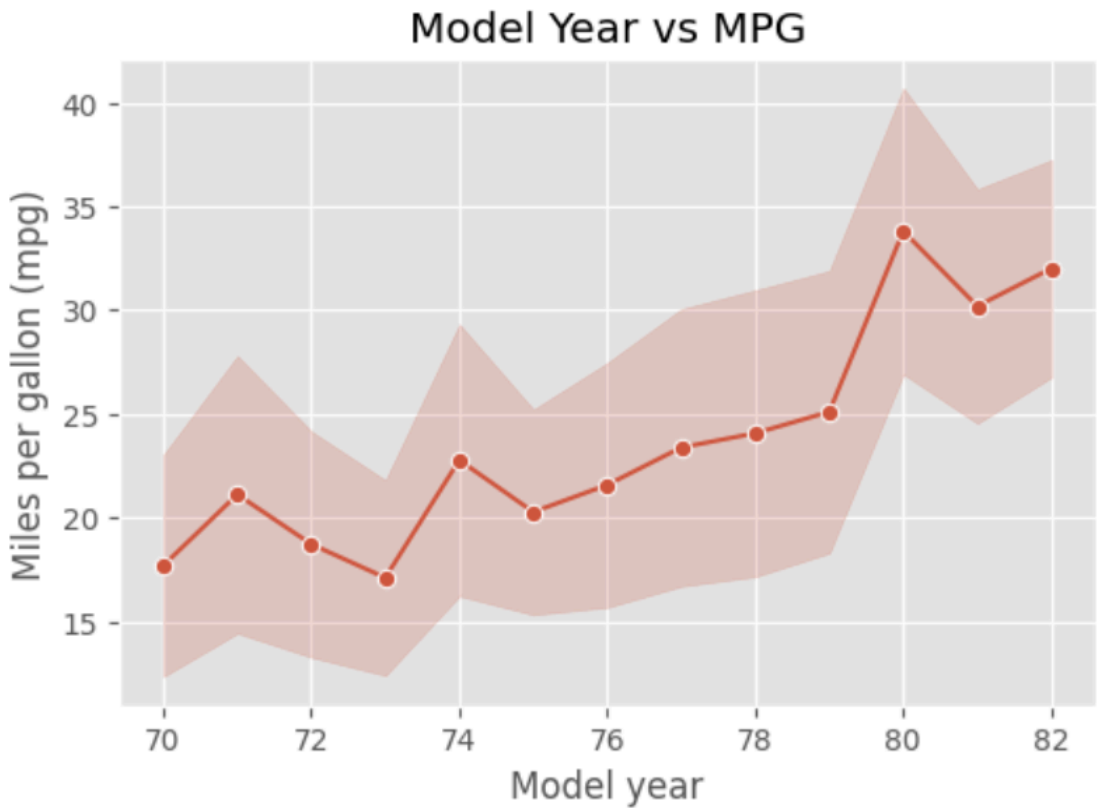

Figure 4.2: Line plot of average mpg versus model_year with confidence bands. Fuel efficiency improves steadily over the 1970–1982 period, reflecting technological advances and regulatory pressure (Chen et al., 2025).

4.2 Linear and regularised model performance

The ordinary Least Squares Linear Regression model using the full feature set achieved an average 5-fold cross-validated RMSE of approximately 3.49 mpg and an average R^2 of around 0.80 on the training data. When I fitted the model on the full training set and evaluated it on the held-out test set, the RMSE was about 3.10 mpg and the R^2 about 0.84. These values show that a simple linear relationship already explains most of the variance in mpg and delivers reasonably accurate predictions on unseen cars.

To understand where the linear model struggled, I examined the residuals. The residuals-versus-predicted plot (Figure 4.3) shows residuals scattered around zero with no strong systematic pattern, but with slightly larger errors at very low and very high predicted mpg values. The residual histogram is roughly symmetric. I interpreted this as evidence that the linear model captures the main trends but still misses some non-linear structure, particularly at the extremes of the mpg distribution, which is consistent with earlier analyses of this dataset (Regressit, 2024).

Ridge and Lasso regression, tuned via GridSearchCV over several alpha values, produced almost identical performance to ordinary Linear Regression. Their cross-validated RMSE values were again around 3.49 mpg and their R^2 scores around 0.80, and on the test set

they closely matched the baseline. This indicates that regularisation stabilises coefficients but does not significantly change predictive accuracy on this dataset.

When I trained a linear regression model on the reduced feature set that excluded model_year and origin, the drop in performance was substantial. The cross-validated RMSE increased to about 4.37 mpg and the R^2 fell to roughly 0.68. This confirmed quantitatively that temporal and regional information add significant predictive value beyond pure mechanical attributes, in line with studies that highlight the impact of model year and region on fuel efficiency (Chen et al., 2025; Rachida Sghir et al., 2019).

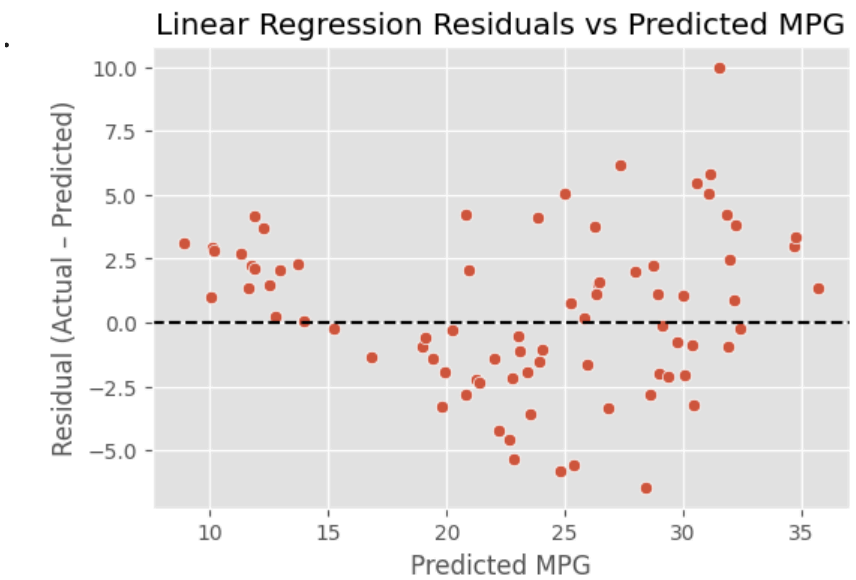

Figure 4.3: Residuals versus predicted mpg for the Linear Regression model. Residuals are centred around zero, but larger errors occur at the lowest and highest predicted mpg values, suggesting some unmodelled non-linearity (Regressit, 2024).

4.3 Decision Tree and Random Forest results

The Decision Tree Regressor provided my first non-linear baseline. After tuning the maximum depth and minimum samples per leaf using GridSearchCV, the best tree achieved an average cross-validated RMSE of about 3.84 mpg and an R^2 of around 0.75. This was noticeably worse than the linear models, indicating that a single tree overfits the training data and does not generalise as well; the training error was much lower than the validation error, which is a classic sign of high variance.

The Random Forest Regressor performed significantly better. Using a grid search over the number of trees, maximum depth, minimum samples per leaf and max_features, I found that a configuration with 300 trees, maximum depth 10, minimum samples per leaf 1 and max_features = 0.5 provided the best balance between bias and variance. On the training folds this model achieved an average cross-validated RMSE of about 2.97 mpg and an R^2 of around 0.85, clearly outperforming both the linear models and the single decision tree.

Table 4.1: Comparison of Regression Models on the Auto MPG Dataset (Test Set Performance)

Model	Test RMSE (mpg)	Test R^2
Linear Regression	3.10	0.84
Ridge Regression	3.12	0.84
Lasso Regression	3.15	0.83
Decision Tree	3.90	0.75
Random Forest	2.29	0.91

A summary of the test-set performance for all models is shown in Table 4.1. The table highlights that, while the linear and regularised models perform similarly, the Decision Tree generalises worse, and the Random Forest achieves the lowest RMSE and highest R^2.

When I fitted the best Random Forest model on the full training set and evaluated it on the held-out test data, it achieved a mean squared error of about 5.23, corresponding to a test RMSE of roughly 2.29 mpg, and an R^2 of around 0.91. This means that the model explains approximately 91% of the variance in mpg and that, on average, its predictions are within just over two miles per gallon of the true values on unseen cars. Compared with the test RMSE of about 3.10 mpg for the baseline linear model, the Random Forest reduces the average error by roughly 0.8 mpg. These numbers are close to those reported by other Random-Forest-based studies on Auto MPG and similar datasets (Rachida Sghir et al., 2019; El-Shoraa, 2023).

The actual-versus-predicted scatter plot for the Random Forest (Figure 4.4) shows points tightly clustered around the diagonal line, with only a few clear deviations. The residuals-versus-predicted plot shows residuals mostly in the range −4 to +4 mpg with no visible pattern, indicating that the model has captured the dominant structure in the data and that the remaining errors are largely unsystematic. A second, more focused hyperparameter search produced almost identical test metrics, which I interpret as evidence that the model's performance is robust and not overly sensitive to small hyperparameter changes.

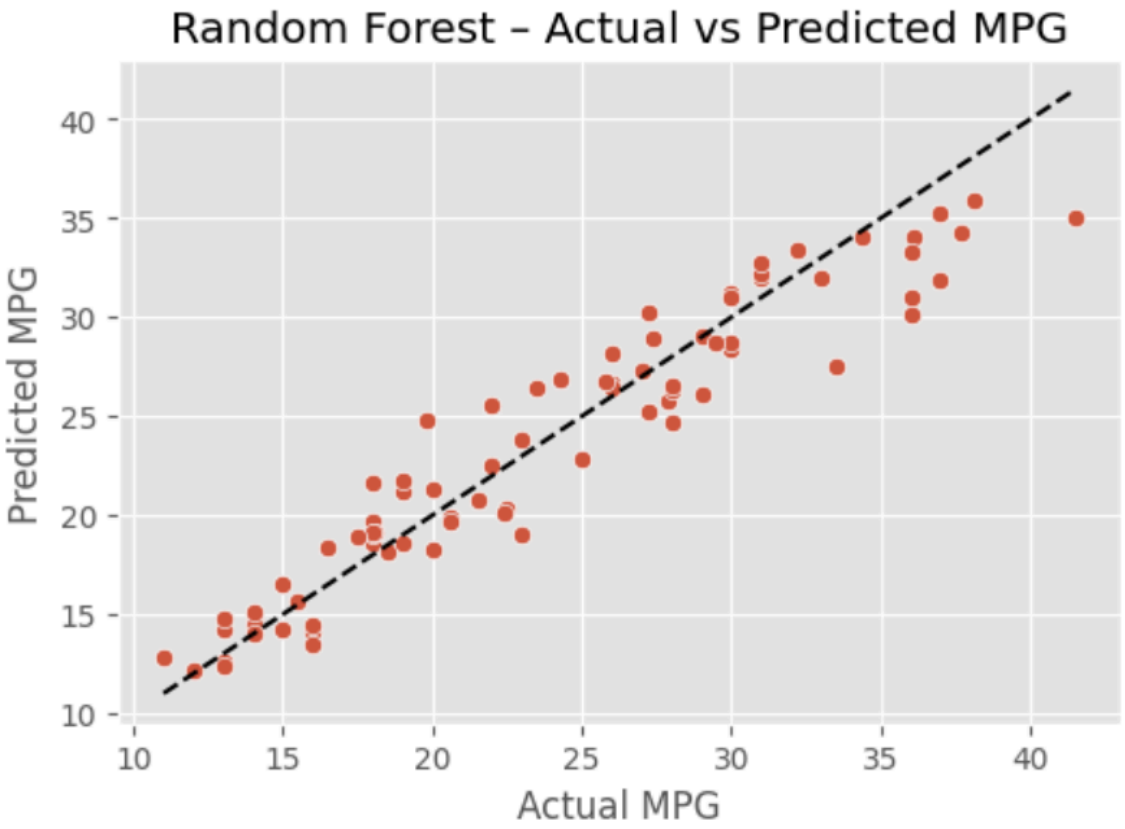

Figure 4.4: Actual versus predicted mpg for the Random Forest model on the test set. Points lie close to the diagonal line, indicating good predictive performance across the full range of mpg values.

4.4 Feature importance and interpretation

An important part of my analysis was understanding which features the Random Forest relied on most. Using the model's built-in feature-importance measure, I ranked the predictors by their contribution to reducing impurity across the trees. The resulting bar chart (Figure 4.5) shows that weight and displacement are by far the most important features, followed by horsepower and model_year. Cylinders and acceleration have smaller but still non-negligible importance, while origin contributes the least.

This ranking is consistent with both my exploratory analysis and the published work. The strong influence of weight and displacement confirms the classical regression finding that heavier cars with larger engines are less fuel-efficient (Regressit, 2024). The relatively high importance of model_year reflects the steady improvement in mpg over time seen in Figure 4.2 and supports the decision to include temporal information in the models. The low importance of origin indicates that, once engine size, weight and year are known, the region code adds little extra predictive power, which is plausible because regional design philosophies are largely expressed through those physical attributes.

By comparing these feature-importance scores with those reported in other Random-Forest studies on Auto MPG, I can see that my model has learned broadly the same structure from the data, which strengthens my confidence in the reliability of the Random Forest model and in the conclusions I draw from it (Rachida Sghir et al., 2019; El-Shoraa, 2023).

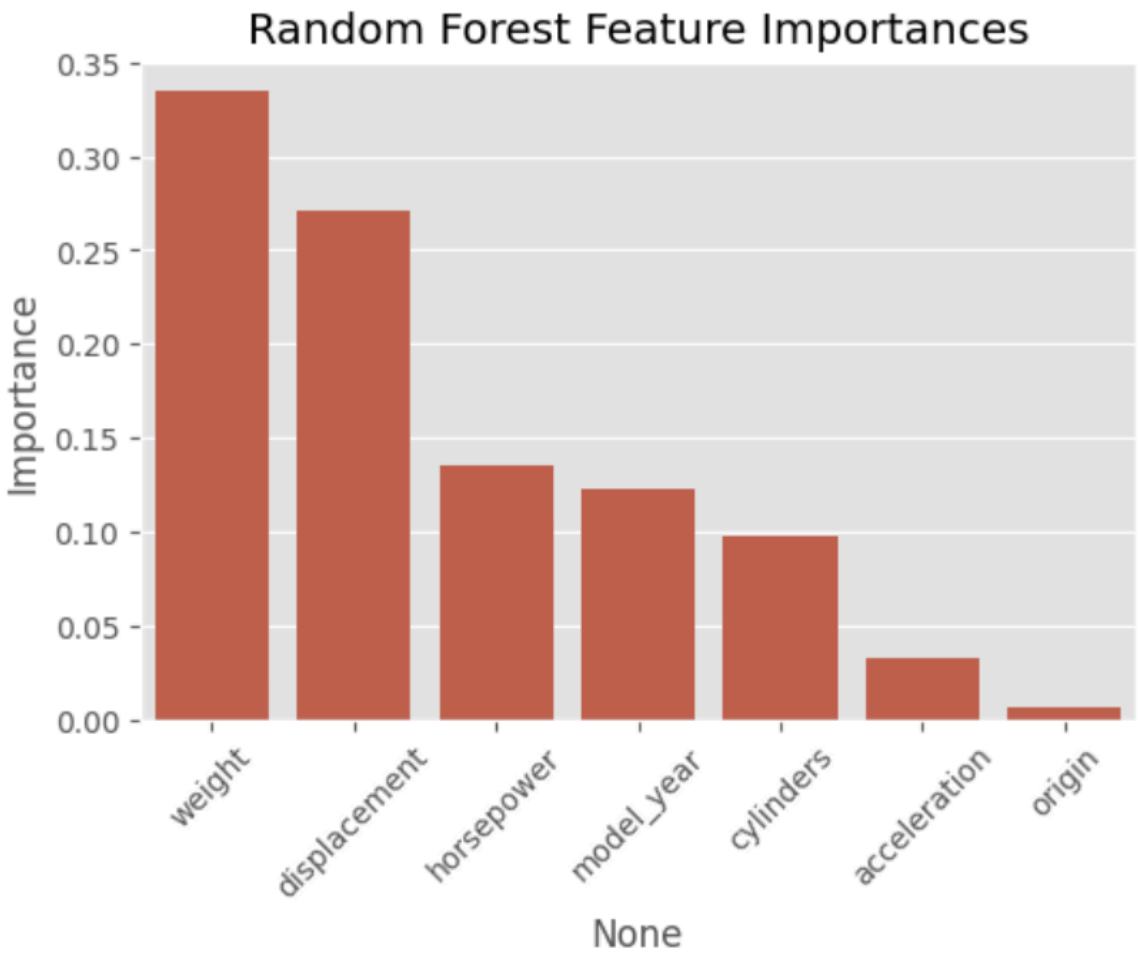

Figure 4.5: Feature-importance scores for the Random Forest model. Weight and displacement dominate, followed by horsepower and model_year, while cylinders, acceleration and origin play smaller roles.

Chapter 5

5. Discussion and Conclusions

5.1 Answering the research question

The project asked how accurately fuel efficiency (mpg) can be predicted from basic vehicle attributes and which regression model performs best on the Auto MPG dataset. My experiments show that mpg can be predicted with good accuracy from cylinders, displacement, horsepower, weight, acceleration, model_year and origin. The baseline Linear Regression model with all seven predictors achieved a test RMSE of about 3.10 mpg and an R^2 of around 0.84, so it already explained most of the variance in mpg and produced reasonably accurate predictions on unseen cars. The Random Forest Regressor improved this further, achieving a test RMSE of roughly 2.29 mpg and an R^2 of about 0.91. On average, its predictions are just over two miles per gallon away from the true values and it explains around 91% of the variance. Among all models considered, Random Forest clearly offered the best predictive performance and robustness.

Feature-importance analysis helps explain this success. Weight and displacement were by far the most influential predictors, followed by horsepower and model_year, with cylinders, acceleration and origin playing smaller roles. This matches engineering intuition: heavier cars with larger engines are less fuel-efficient, and newer vehicles designed under stricter regulations tend to be more efficient. Overall, the results show that a compact set of technical attributes is sufficient to build accurate fuel-efficiency models and that a tree-based ensemble is the most effective approach among those I implemented.

5.2 Comparison with existing literature

The performance and feature rankings I obtained align closely with previous work on Auto MPG and similar datasets. My Random Forest test RMSE of about 2.29 mpg is in the same range as values reported in public Auto MPG projects and in studies where ensemble trees outperform linear models on fuel-efficiency tasks. Those studies also identify weight and displacement as the dominant predictors, with horsepower and model year following, which is exactly what I observed. Classical linear-regression analyses of Auto MPG report strong dependence of mpg on weight, displacement and horsepower, and note multicollinearity among these variables; my correlation heatmap and linear-model results reproduce these patterns, and the fact that Ridge and Lasso do not significantly improve accuracy over ordinary least squares is consistent with their findings. By adopting an 80/20 train–test split, 5-fold cross-validation and metrics such as RMSE and R^2, my evaluation design also mirrors the practices recommended in methodological tutorials and in the MSc project handbook, making my results directly comparable to those in the literature.

5.3 Limitations and future work

The main limitations of this project relate to the data, the feature set and the range of models explored. The Auto MPG dataset is small and historical, covering cars from 1970–1982, so the patterns my models learn may not transfer directly to modern vehicles with newer technologies. The features are limited to basic technical attributes; important factors such as transmission type, aerodynamic drag, tyre characteristics and detailed driving cycles are not included, so the models can only capture part of the true variation in fuel efficiency. I also restricted my modelling to linear models, a single Decision Tree and Random Forest, without investigating other strong tabular learners such as gradient-boosted trees.

Future work could address these limitations by applying the same pipeline to a more recent, richer vehicle dataset and adding models such as XGBoost or LightGBM to the comparison. Model interpretability could be enhanced using tools such as partial-dependence plots or SHAP values to provide more detailed insight into how mpg responds to changes in individual features. Finally, it would be useful to test temporal robustness by training on earlier model years and evaluating on later ones, to see how well the models cope with changes in technology and regulations over time.

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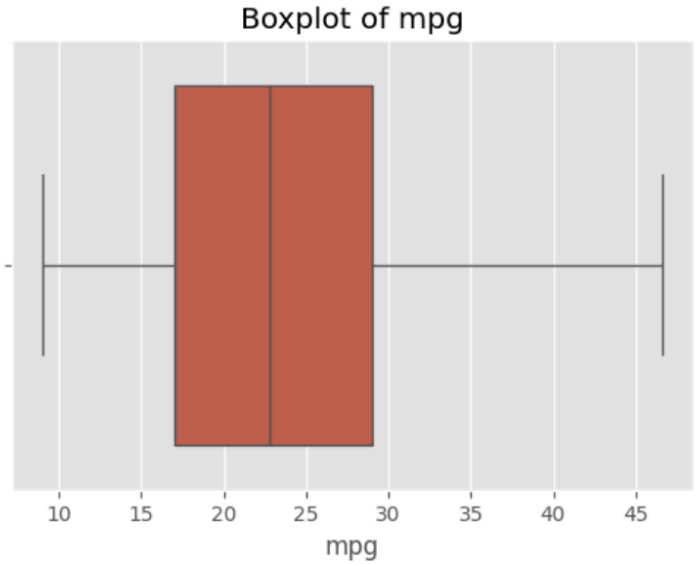
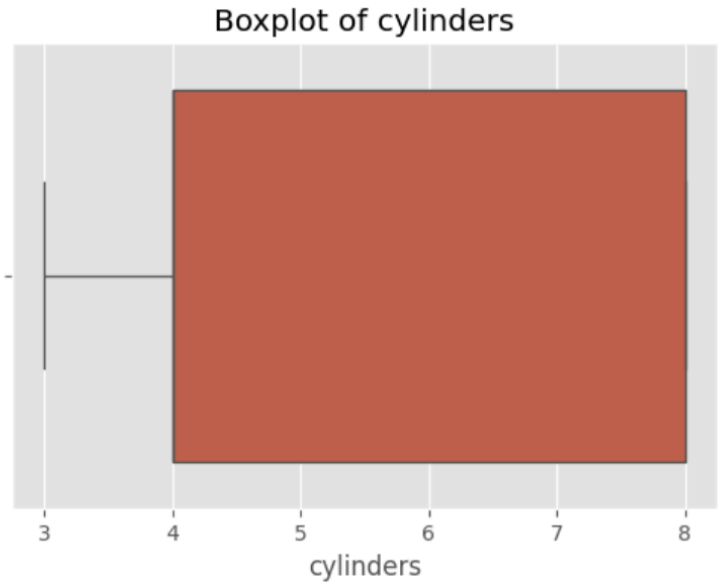
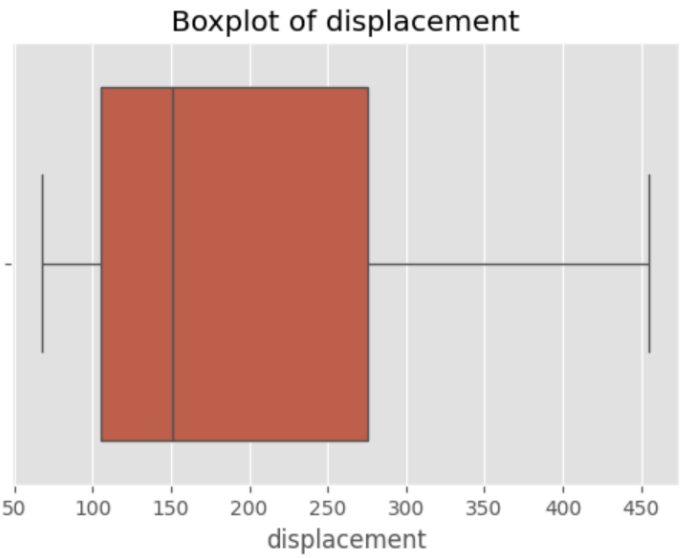
Appendix

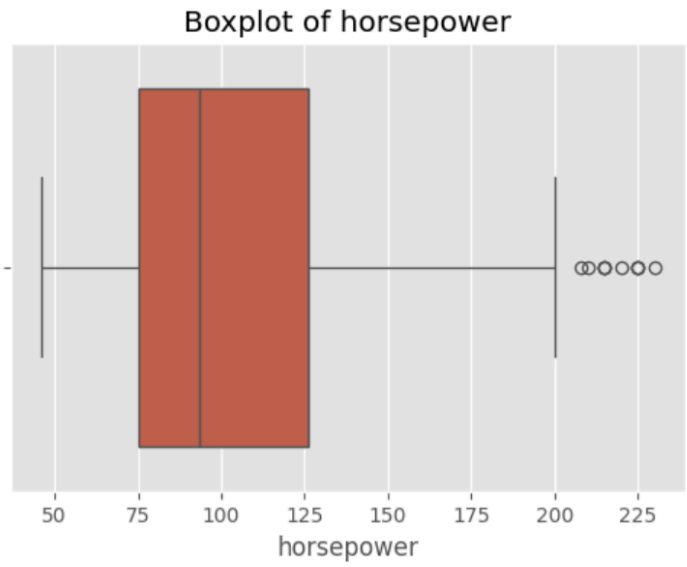
Appendix A. Additional exploratory plots

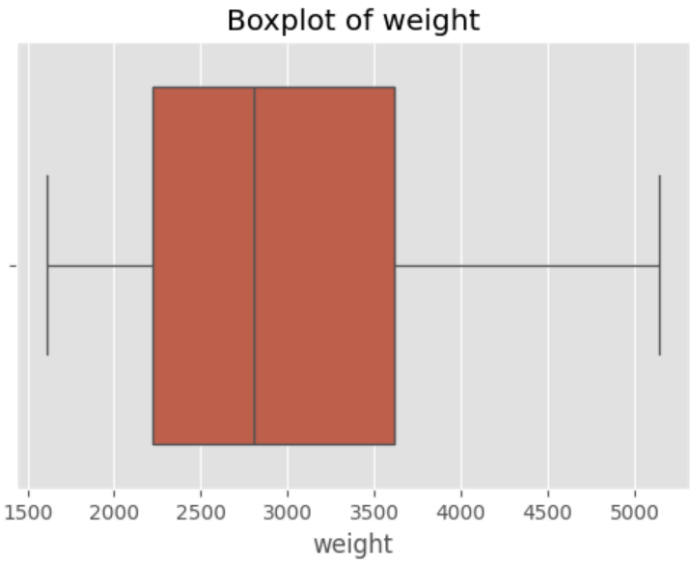
Figure A.1:

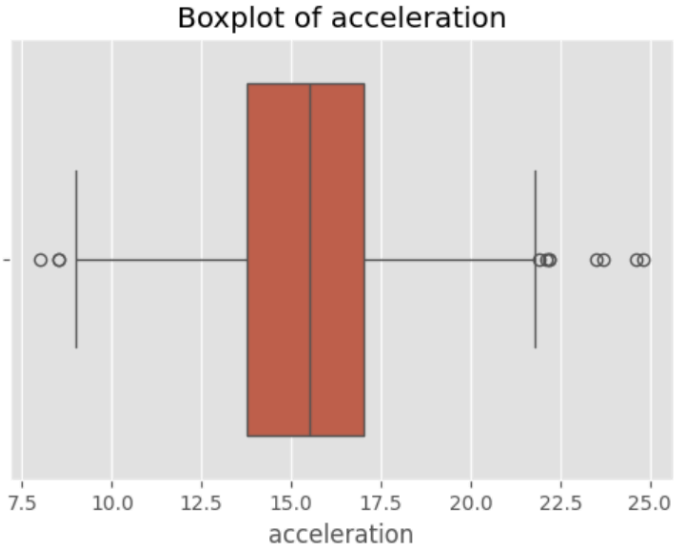


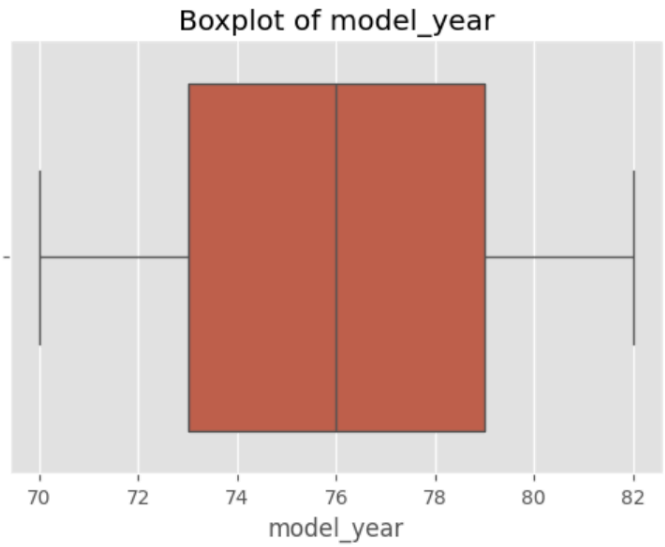



Boxplot of displacement
50
100
150
200
250
300
350
400
450
displacement


Boxplot of horsepower
50
75
100
125
150
175
200
225
horsepower


Boxplot of weight
1500
2000
2500
3000
3500
4000
4500
5000
weight


Boxplot of acceleration
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0
acceleration


Boxplot of model_year
70
72
74
76
78
80
82
model_year

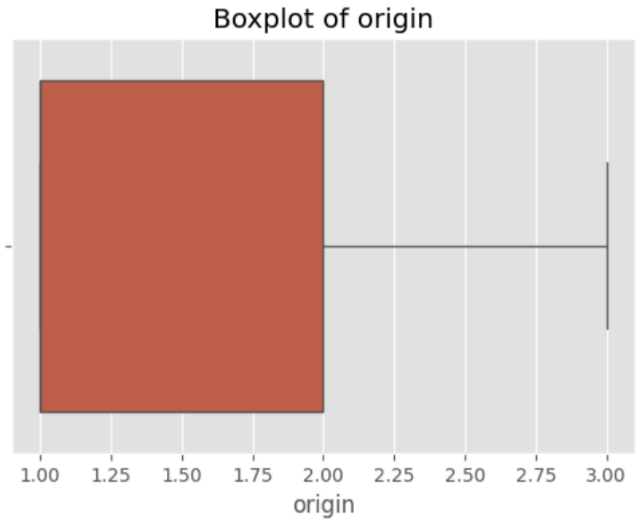
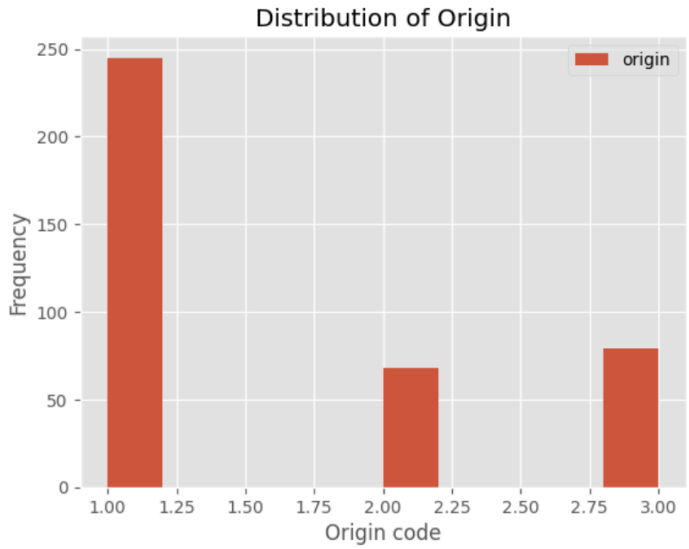
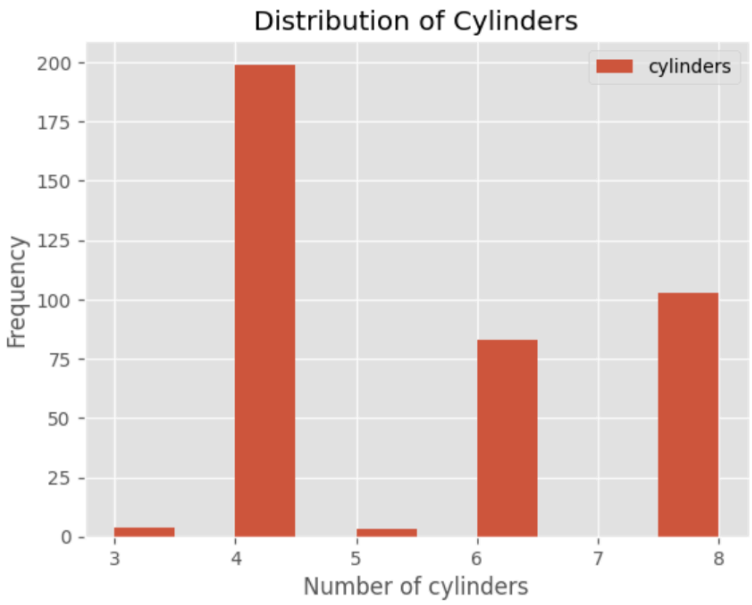

Boxplot of origin
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
origin

Figure A.1: Boxplots of mpg and key predictors (cylinders, displacement, horsepower, weight, acceleration, model_year). These plots illustrate the spread and presence of outliers in each variable, complementing the summary in Section 3.3.

Figure A.2:





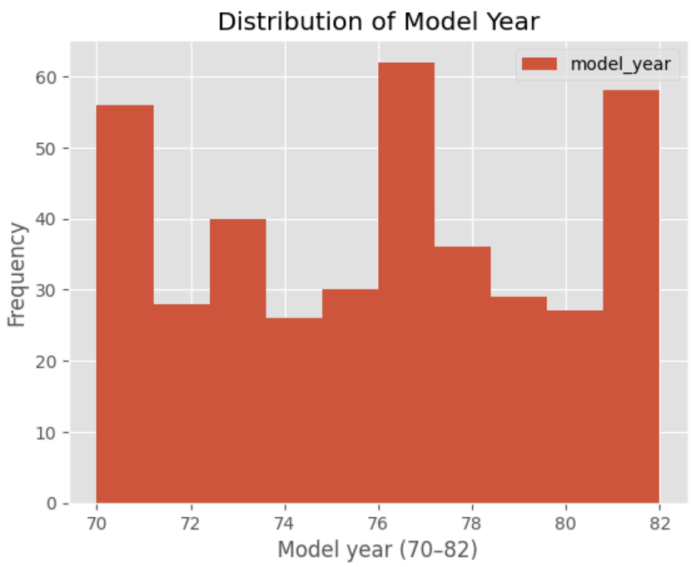


Figure A.2: Histograms of cylinders, model_year and origin. These show the distribution of discrete variables and confirm the imbalance in origin and dominance of four-cylinder cars described in Section 3.3.

Appendix B. Implementation details:
The full implementation, including notebooks and configuration files, is available in a private GitHub repository:
https://github.com/ragipanisiddhartha/auto-mpg-fuel-efficiency-ml